0.1.1 Welcome

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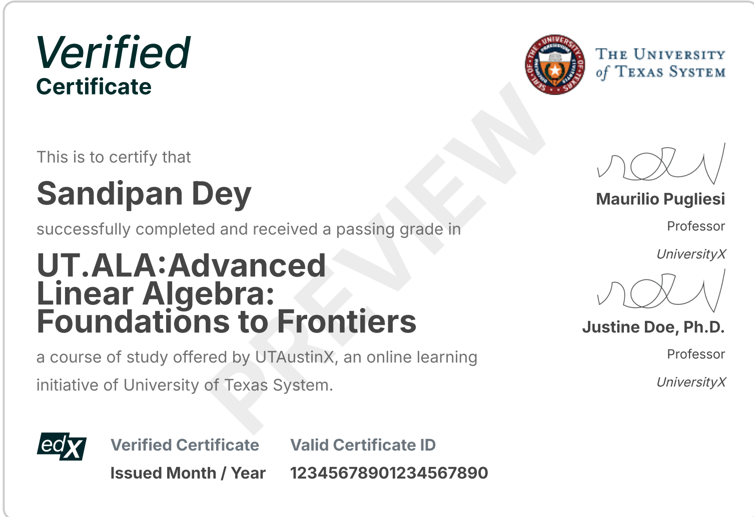

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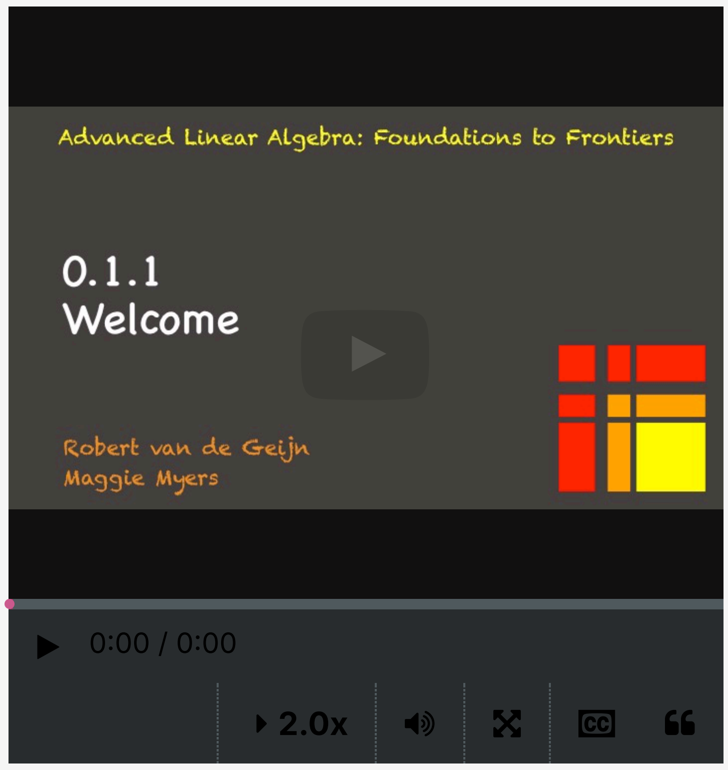

Start of transcript. Skip to the end.

On behalf of Maggie, who is notoriously camera shy, and myself, we welcome you to this course.

You should call us Robert and Maggie, since we are very informal around here.

Linear algebra is fundamental in this age of technology.

Video

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Reading assignment

0 points possible (ungraded)

Read Unit 0.1.1 of the notes. [LINK]

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Huge thanks to both Prof. Geijn and Myers
I can't say enough to thank you both for the great course (the basic LAFF c... 5

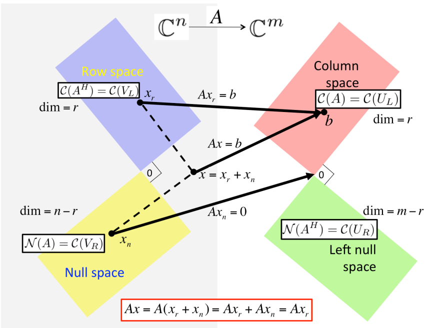
Huge Shoutout To Professor Robert & Dr. Maggie :)
Thanks to both of you so much for the amazing work and service that you p... 4

Linear algebra is one of the fundamental tools for computational and data scientists. In Advanced Linear Algebra: Foundations to Frontiers (ALAFF), you build your knowledge, understanding, and skills in linear algebra, practical algorithms for matrix computations, and how floating-point arithmetic as performed by computers affects correctness.

Organization of this course

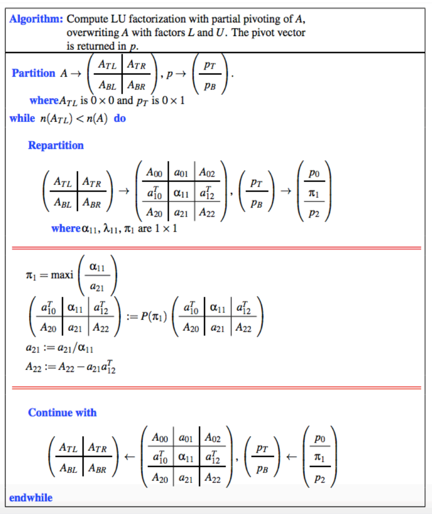
The materials are organized into Weeks that correspond to a chunk of information that is covered in a typical on-campus week. These weeks are arranged into three parts. The topics are ordered so that we get to the very important Singular Value and QR Decompositions early on. Each part will be challenging, but worth it!

Part I: Orthogonality



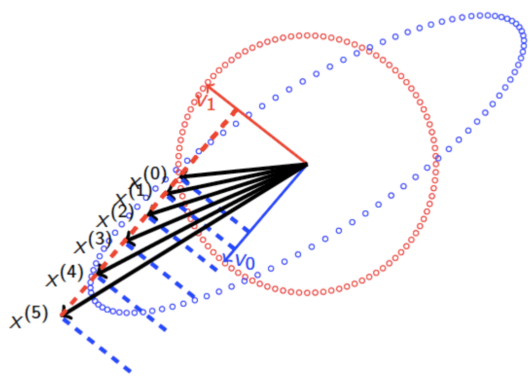
The Singular Value Decomposition (SVD) is possibly the most important result in linear algebra, yet too advanced to cover in an introductory undergraduate course. To be able to get to this topic as quickly as possible, we start by focusing on orthogonality, which is at the heart of image compression, Google's page rank algorithm, and linear least-squares approximation.

Part II: Solving Linear Systems



Solving linear systems, via direct or iterative methods, is at the core of applications in computational science and machine learning. We also leverage these topics to introduce numerical stability of algorithms: the classical study that qualifies and quantifies the "correctness" of an algorithm in the presence of floating point computation and approximation. Along the way, we discuss how to restructure algorithms so that they can attain high performance on modern CPUs.

Part III: Eigenvalues and Eigenvectors



Many problems in science have the property that if one looks at them in just the right way (in the right basis), they greatly simplify and/or decouple into simpler subproblems. Eigenvalue and eigenvectors are the key to discovering how to view a linear transformation, represented by a matrix, in that special way. Algorithms for computing them also are the key to practical algorithms for computing the SVD.

In this week (Week 0), we walk you through some of the basic course information and help you set up for learning. The week itself is structured like future weeks, so that you become familiar with that structure.

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